

severe environment where protective earth may already be polluted. If protective earth has to be extended to the modules via, say, a dual-in-line (DIN) rail-mounted terminal block, then it can be done by a separate wire to comply with requirements of electrical safety standards.

Earthing can be extended to hinged doors by using bond straps (braided or sheet metal). For more effective earthing of metal doors, EMI gaskets can be installed over the mating surface at the periphery.

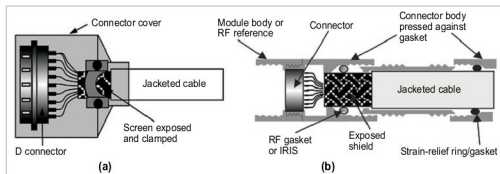


Fig. 26: Bonding of shield to connector

Bonding screened cables with connectors. Screened cables are many a time used to bring signals from the external world to modules inside a cabinet. When these enter the cabinet, either through a connector or otherwise, their shields need to be connected to the local RF reference (which can be the mounting plate or metal enclosure of the module, or even the PCB reference plane) through a low-RF impedance bond.

When shielded connectors with metal or metal covers such as D-type connectors are used, cable shields need to be bonded to the connector body. The simplest way is to use a semi-circular or saddle clamp. As shown in Fig. 26 (a), the shield is exposed in such a way that the exposed surface area is just a shade greater than the clamp surface area.

Care should be taken that curvature of the clamp matches that of the cable to ensure total contact

over the entire mating surface. The clamp, when tightened, exerts a small pressure over the shield. The clamp is then fixed on the connector shell at two points. Since the saddle clamp is semi-circular, it may not provide full 360° and, consequently, the connector can be used in situations where shielding-effectiveness requirements are not that high.

The best way to have a true 360° contact as well as to provide good strain relief is to use a circular metallic connector, where the shield

the shield be bonded to the mounting plate (RF reference) at a point closest to the module or even to the module body itself (which is, in turn, bonded to the mounting plate). There are various methods suggested by manufacturers for bonding the shield to RF reference. Those methods where the exposed shield is bonded to the reference by a separate wire or where shield itself is twisted and bonded to the reference (called pigtailed) are generally not recommended since their RF performance is dismal.

Pigtails are typically worse by more than 35dB for frequencies above a few megahertz than other methods. A better method is to employ P-clips, which clamp around the exposed shield and bond to mounting plate (or module body) at a single point. Saddle clamps provide an even better method where the shield

is bonded to the clamp, which is, in turn, bonded to the mounting plate at two points.

Conclusion

In the previous and current parts of the article, we have seen various methods of grounding and bonding. These methods are not only important themselves but also contribute in a big way in increasing the efficiency of other design methods to ensure EMC like filtering and shielding.

There is no denying the fact that these measures have to be taken into consideration before going for fixes like filtering and shielding as these are design measures and do not cost much, if taken care of before freezing equipment design. These can further reduce cost of filters and shielding measures by reducing their performance requirements and thus contribute in reducing the overall cost of the equipment.

To be continued...